

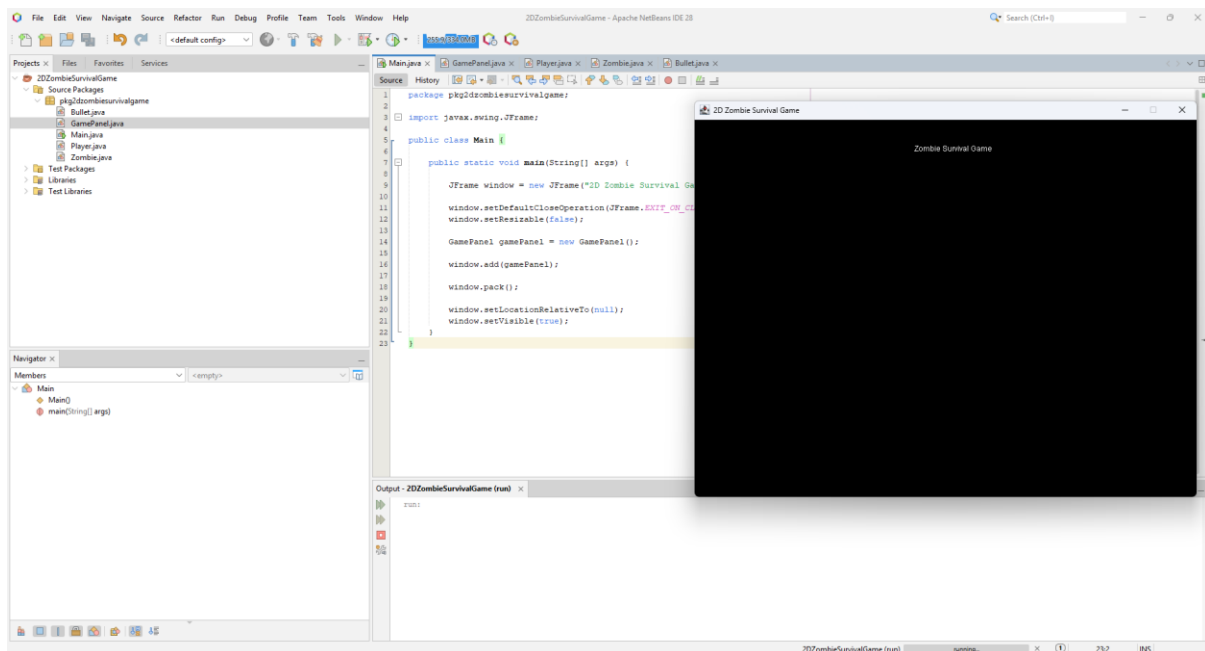
Week 3: Game Design & Project Setup

Activities Performed:

- Designed the basic layout of the game window.
- Decided the positions and interactions of the player, zombies, bullets, score display, and health bar.
- Created the Java project named "Zombie Survival Game".
- Created the main classes required for development:
 - ❖ Main.java
 - ❖ GamePanel.java
 - ❖ Player.java
 - ❖ Zombie.java
 - ❖ Bullet.java
- Implemented the game window using JFrame.
- Added a custom GamePanel using JPanel.
- Successfully tested the project execution and verified that the game window opens correctly.

Outcome:

The project structure has been completed successfully. The game window is functional and ready for implementing the player character and game mechanics in the next phase.



All Codes :

➤ Main.java Code:

```
package pkg2dzombiesurvivalgame;
import javax.swing.JFrame;
public class Main {
    public static void main(String[] args) {
        JFrame window = new JFrame("2D Zombie Survival Game");
        window.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        window.setResizable(false);
        GamePanel gamePanel = new GamePanel();
        window.add(gamePanel);
        window.pack();
        window.setLocationRelativeTo(null);
        window.setVisible(true);
    }
}
```

➤ GamePanel.java Code:

```
package pkg2dzombiesurvivalgame;
import javax.swing.JPanel;
import java.awt.Color;
import java.awt.Dimension;
import java.awt.Graphics;
public class GamePanel extends JPanel {
    final int screenWidth = 800;
    final int screenHeight = 600;
    public GamePanel() {
        this.setPreferredSize(
            new Dimension(screenWidth, screenHeight));
        this.setBackground(Color.BLACK);
        this.setDoubleBuffered(true);
    }
    @Override
    protected void paintComponent(Graphics g) {
        super.paintComponent(g);
        g.setColor(Color.WHITE);
        g.drawString("Zombie Survival Game", 350, 50);
        g.dispose();
    }
}
```

➤ **Player.java Code:**

```
package pkg2dzombiesurvivalgame;
public class Player {
    int x;
    int y;
    int health;
    int speed;
    public Player() {
        x = 400;
        y = 300;
        health = 100;
        speed = 5;
    }
}
```

➤ **Zombie.java Code:**

```
package pkg2dzombiesurvivalgame;
public class Zombie {
    int x;
    int y;
    int speed;
    public Zombie(int x, int y) {
        this.x = x;
        this.y = y;
        speed = 2;
    }
}
```

➤ **Bullet.java Code:**

```
package pkg2dzombiesurvivalgame;
public class Bullet {
    int x;
    int y;
    int speed;
    public Bullet(int x, int y) {
        this.x = x;
        this.y = y;
        speed = 8;
    }
}
```